

RaceDay System – API Endpoint Plan

This document lists every API endpoint the RaceDay system will expose, planned before any API code is written, in line with Part 1 of the Portfolio of Evidence. Two roles are used throughout: **Member** (a registered participant who can browse and enrol in events) and **Organiser** (an administrator who manages categories, events and results). Sections 7 and 8 are additional endpoints identified beyond the minimum requirement: exposing an event's location for map display, and wrapping an external weather API to show event-day forecasts.

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Creates a new user account	None (public)	{ fullName, email, password, roleId }	201 Created – new user record 400 Bad Request – missing/invalid fields 409 Conflict – email already registered
POST	/api/auth/login	Verifies credentials and issues an access token	None (public)	{ email, password }	200 OK – JWT token + user info 401 Unauthorized – wrong email or password

2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the logged-in user's own profile details	Any (logged in)	None	200 OK – user's name, email, role 401 Unauthorized – not logged in
PUT	/api/users/me	Updates the logged-in user's own profile	Any (logged in)	{ fullName, email }	200 OK – updated profile 400 Bad Request – invalid data
GET	/api/users	Lists all registered users (admin overview)	Organiser	None	200 OK – array of users 403 Forbidden – not an organiser

3. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/categories	Lists all available event categories	None (public)	None	200 OK – array of categories
POST	/api/categories	Creates a new event category	Organiser	{ categoryName }	201 Created – new category 403 Forbidden – not an organiser
PUT	/api/categories/{id}	Updates an existing category's name	Organiser	{ categoryName }	200 OK – updated category 404 Not Found – category doesn't exist

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
DELETE	/api/categories/{id}	Removes a category	Organiser	None	204 No Content – deleted 404 Not Found – category doesn't exist

4. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists all upcoming events, optionally filtered by category	None (public)	None	200 OK – array of events
GET	/api/events/{id}	Returns full details for a single event	None (public)	None	200 OK – event details 404 Not Found – event doesn't exist
POST	/api/events	Creates a new event	Organiser	{ eventName, eventDate, location, categoryId }	201 Created – new event 400 Bad Request – invalid data 403 Forbidden – not an organiser
PUT	/api/events/{id}	Updates an existing event's details	Organiser	{ eventName, eventDate, location, categoryId }	200 OK – updated event 404 Not Found – event doesn't exist
DELETE	/api/events/{id}	Cancels/removes an event	Organiser	None	204 No Content – deleted 404 Not Found – event doesn't exist

5. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{id}/enrol	Enrols the logged-in user in a specific event	Member	None	201 Created – enrolment record 404 Not Found – event doesn't exist 409 Conflict – already enrolled
GET	/api/users/me/enrolments	Lists all events the logged-in user has enrolled in	Member	None	200 OK – array of enrolments
DELETE	/api/enrolments/{id}	Cancels the user's own enrolment	Member	None	204 No Content – cancelled 403 Forbidden – not your enrolment 404 Not Found – enrolment doesn't exist
GET	/api/events/{id}/enrolments	Lists everyone enrolled in a specific event	Organiser	None	200 OK – array of enrolled users 404 Not Found – event doesn't exist

6. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{id}/results	Captures the result for a specific enrolment	Organiser	{ finishTime, position }	201 Created – result record 404 Not Found – enrolment doesn't exist 409 Conflict – result already captured
GET	/api/events/{id}/results	Lists all results for a specific event (leaderboard)	None (public)	None	200 OK – array of results ordered by position
GET	/api/users/me/results	Lists the logged-in user's own race history and results	Member	None	200 OK – array of the user's past results

7. Location (additional endpoint)

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}/location	Returns an event's location details (address, latitude, longitude) for display on a map	None (public)	None	200 OK – location name, latitude, longitude 404 Not Found – event doesn't exist

8. Weather (additional endpoint)

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}/weather	Fetches the weather forecast for an event's location and date by calling an external weather provider server-side, using the event's stored latitude/longitude	None (public)	None	200 OK – forecast (temperature, conditions, wind) 404 Not Found – event doesn't exist 502 Bad Gateway – weather provider unavailable